INDIVIDUAL TASK-01

Working of Artificial Neural Networks

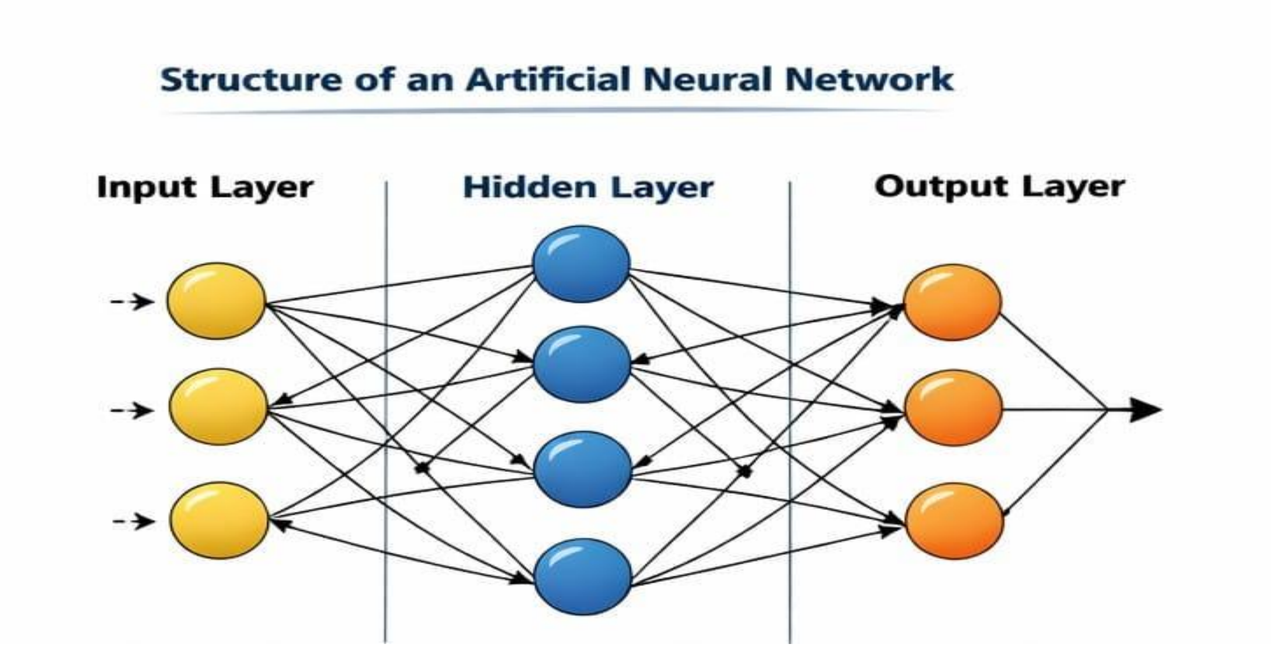
1. Introduction

Artificial Neural Networks (ANNs) are computational models inspired by the biological neural network of the human brain. The human brain consists of billions of interconnected neurons that communicate through electrical and chemical signals. Similarly, ANNs consist of artificial neurons connected in layers to process information.

Artificial Neural Networks are a major part of Artificial Intelligence and Machine Learning. They are designed to recognize patterns, classify data, and make predictions by learning from past experiences. Unlike traditional programming, where instructions are explicitly written, neural networks learn automatically from data.

ANNs are widely used in modern technologies such as image recognition, speech processing, medical diagnosis, financial forecasting, and autonomous systems.

2. Basic Structure of Artificial Neural Networks



An Artificial Neural Network is composed of multiple layers of interconnected neurons. These layers are arranged in a structured manner to process input data and generate output.

2.1 Input Layer

The input layer is the first layer of the neural network. It receives raw data from the external environment. Each neuron in this layer represents one feature of the input data.

For example, in a student performance prediction system, the input features may include attendance percentage, study hours, and previous examination marks.

2.2 Hidden Layer

Hidden layers are located between the input and output layers. These layers perform complex mathematical operations on the input data. The hidden layer extracts meaningful patterns and relationships from the data.

A neural network may contain one or more hidden layers. Networks with multiple hidden layers are called Deep Neural Networks.

2.3 Output Layer

The output layer produces the final result of the network. The number of neurons in this layer depends on the type of problem.

For example:

- One neuron for binary classification (Yes/No)
- Multiple neurons for multi-class classification

3. Components of Artificial Neural Network

Each artificial neuron consists of the following components:

Inputs (X_1, X_2, X_3, …)

These are the signals received from the previous layer or external data.

Weights (W_1, W_2, W_3, …)

Weights determine the importance of each input. Higher weight indicates stronger influence on the output.

Bias (b)

Bias is an additional parameter that helps improve the flexibility and accuracy of the model.

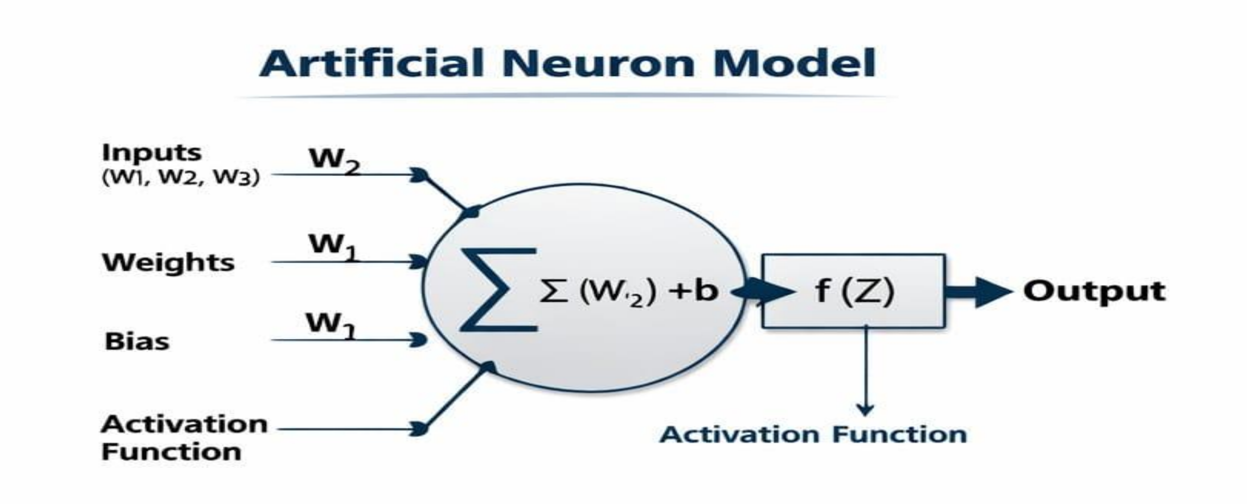
Activation Function

The activation function determines whether a neuron should activate or not. It introduces non-linearity into the system.

Common activation functions include:

- Sigmoid Function
- ReLU (Rectified Linear Unit)
- Tanh Function
- Softmax Function

4. Mathematical Model of Artificial Neuron



The artificial neuron performs the following operation:

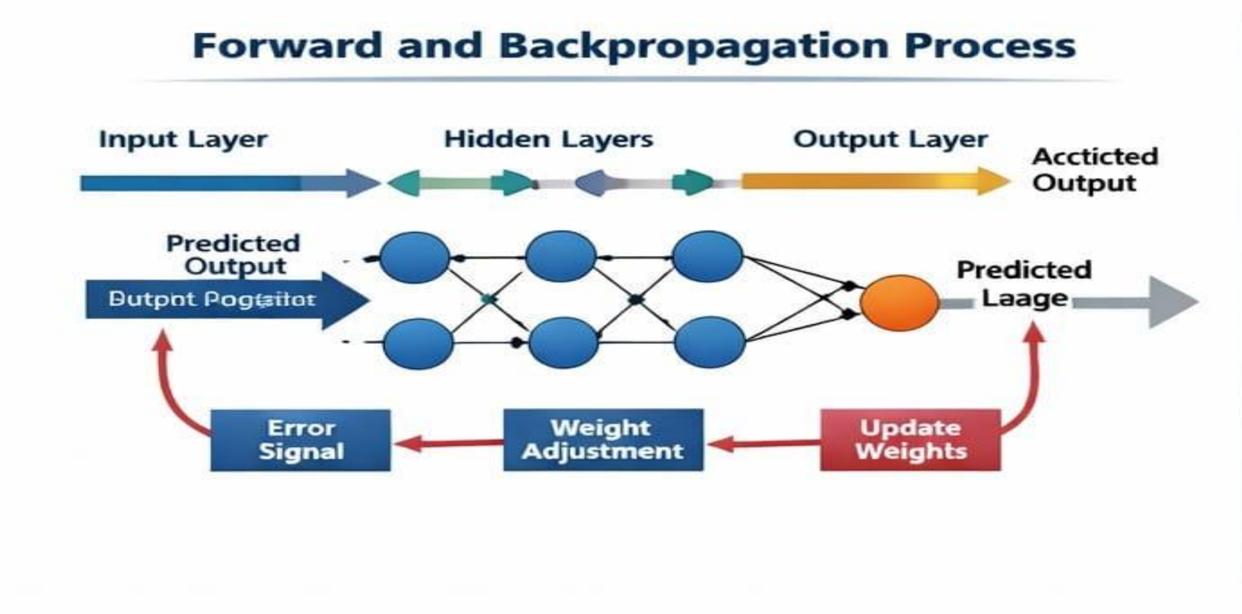
The output is calculated as:

Where:

- Wi represents weights
- Xi represents inputs
- b represents bias
- f(Z) represents activation function

This mathematical process allows the network to convert input data into meaningful output.

5. Working of Artificial Neural Networks:



The working of ANN mainly involves two important phases:

5.1 Forward Propagation

In forward propagation, data moves from the input layer to the output layer.

1. Input data is provided to the input layer.
2. Each neuron calculates the weighted sum of inputs.
3. The activation function is applied.
4. The output is passed to the next layer.
5. Final prediction is generated at the output layer.

This process produces the predicted output based on current weight values.

5.2 Error Calculation

After obtaining the predicted output, it is compared with the actual output. The difference between them is called error.

Error is measured using loss functions such as:

- Mean Squared Error (MSE)
- Cross Entropy Loss

The goal of the network is to minimize this error.

5.3 Backpropagation

Backpropagation is the process of reducing error by adjusting weights.

1. The error is propagated backward through the network.
2. Partial derivatives are calculated.
3. Weights and bias values are updated.
4. The network gradually improves its accuracy.

Backpropagation uses an optimization algorithm called Gradient Descent to minimize the loss function.

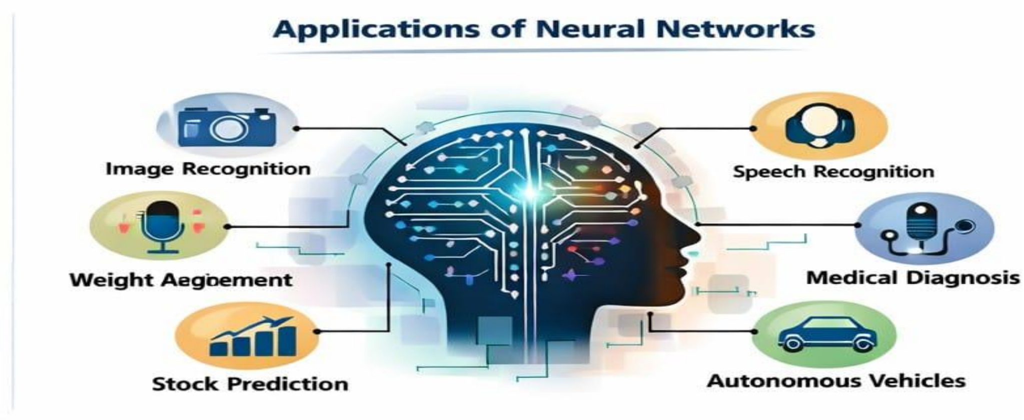
6. Training of Artificial Neural Networks

Training is the process of teaching the neural network using a dataset. During training:

- Large amounts of labeled data are used.
- The network performs multiple iterations known as epochs.
- Weights are continuously updated.

Training continues until the error becomes sufficiently small and acceptable accuracy is achieved.

6. **Applications of Artificial Neural Networks**



Artificial Neural Networks are used in various fields:

Healthcare :For disease prediction and medical image analysis.

Finance : For fraud detection and stock market analysis.

Transportation : For self-driving cars and traffic management.

Communication : For speech recognition and language translation.

Education : For intelligent tutoring systems.

8. Advantages of Artificial Neural Networks

- Ability to learn complex patterns
- High accuracy in prediction
- Self-adaptive system
- Works efficiently with large datasets

9. Limitations of Artificial Neural Networks

- Requires large training data
- High computational power needed
- Difficult to interpret internal working
- Training process may take long time

10. Conclusion

Artificial Neural Networks are powerful computational models that simulate the working of the human brain. They process input data through multiple layers, adjust weights using backpropagation, and continuously improve their performance through training.

With the rapid advancement of Artificial Intelligence, neural networks are becoming increasingly important in solving real-world problems. They play a crucial role in modern technology and are expected to contribute significantly to future innovations.